

Database Management Completed SPARK Reference

HarbourView Community Library - question-by-question completed guide

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: HarbourView Community Library

Sector: Education

Questions being answered

Task A - Tables

1. Create a database named HarbourViewLibrary_YourFullName.accdb.

How SPARK completed it

- Open your database software and create a blank database.
- Save it immediately with the exact filename in the question.
- Keep the database and all later SBA files inside one organised project folder.

Completed result

SPARK reference filename: HarbourViewLibrary_YourFullName.accdb.

2. Create the Member, Book and Loan tables using suitable field names, data types and primary keys.

How SPARK completed it

- Create Member first and enter the fields exactly as planned in your data dictionary.
- Repeat the process for Book and Loan.
- Choose the primary key before entering records. Check currency, date and numeric data types carefully.

Completed result

Completed table structures:

Member: MemberID - Short Text, primary key, FirstName - Short Text, LastName - Short Text, MemberType - Short Text, Phone - Short Text, Email - Short Text

Book: BookID - Short Text, primary key, Title - Short Text, Category - Short Text, Author - Short Text, DailyFine - Currency

Loan: LoanID - Short Text, primary key, MemberID - Short Text, foreign key, BookID - Short Text, foreign key, DateBorrowed - Date/Time, DueDate - Date/Time, DateReturned - Date/Time

3. Enter realistic records and apply the validation rules required by the project.

How SPARK completed it

- Enter enough varied records to make your queries meaningful.
- Use validation rules for fields with a limited set or range of allowed values.
- Test the validation by deliberately entering one invalid value and confirming that the software rejects it.

Completed result

The SPARK completed database uses fictional records with different categories, dates, fees and statuses so every query can be tested.

4. Create the required relationships and enforce the correct links between the tables.

How SPARK completed it

- Open the Relationships window.
- Add all tables and drag each primary key to the matching foreign key.
- Check the real-world relationship before selecting referential-integrity options.

Completed result

Member.MemberID 1-to-many Loan.MemberID; Book.BookID 1-to-many Loan.BookID.

Task B - Queries and Form

1. Create a query called Overdue Loans. Show loans returned after the due date or still overdue. Include member, book and days late.

How SPARK completed it

- Add only the table or tables required by the question.
- Add the requested fields to the query grid.
- Enter the criterion: FineDue: [DaysLate]*[DailyFine].
- Run the query and compare every returned record with the source data.

Completed result

Query name: Overdue Loans. Criterion used: FineDue: [DaysLate]*[DailyFine].

2. Create a query called Member Loans. Join Member, Book and Loan and show all borrowing details sorted by LastName.

How SPARK completed it

- Add the related tables and confirm the join lines are correct.
- Add the display fields requested in the question.
- Create the calculated field using: No additional calculation.
- Run the query and calculate one record manually to verify the result.

Completed result

Query name: Member Loans. Calculation used: No additional calculation.

3. Create Member Information Form with Loan subform and clear borrowing headings.

How SPARK completed it

- Use the main record source for the parent form.
- Use the related table as the subform source.
- Add a clear title, useful instructions and navigation controls.
- Open several records and confirm the subform changes with the main record.

Completed result

Completed form design: Member Information Form with Loan subform and clear borrowing headings.

Task C - Report

1. Create Overdue Report grouped by MemberType and showing count and total FineDue.

How SPARK completed it

- Use a query as the report source when it already contains the correct fields or calculation.

- Apply the exact grouping and sorting required by the question.
- Add the required summary in the group or report footer.
- Preview the report and correct clipped headings, blank pages and unreadable columns.

Completed result

Completed report: Overdue Report grouped by MemberType and showing count and total FineDue.

SPARK REFERENCE - DO NOT SUBMIT

SPARK REFERENCE - DO NOT SUBMIT

SPARK REFERENCE - DO NOT SUBMIT

SPARK REFERENCE - DO NOT SUBMIT

Completed database structure

Member

Field	Design
MemberID	Short Text, primary key
FirstName	Short Text
LastName	Short Text
MemberType	Short Text
Phone	Short Text
Email	Short Text

Book

Field	Design
BookID	Short Text, primary key
Title	Short Text
Category	Short Text
Author	Short Text
DailyFine	Currency

Loan

Field	Design
LoanID	Short Text, primary key
MemberID	Short Text, foreign key
BookID	Short Text, foreign key
DateBorrowed	Date/Time
DueDate	Date/Time
DateReturned	Date/Time

Relationships

Member.MemberID 1-to-many Loan.MemberID; Book.BookID 1-to-many Loan.BookID.

Sample records

LoanID	MemberName	MemberType	BookTitle	Category	DaysLate	DailyFine
L001	Janelle Rose	STUDENT	Caribbean Voices	Literature	3	100
L002	Paul Green	ADULT	Excel Basics	Technology	5	100
L003	Kemar White	STUDENT	Island History	History	0	100
L004	Sasha Morgan	STUDENT	Physics Made Clear	Science	2	100
L005	Denise Brown	ADULT	Healthy Living	Health	7	100
L006	Andre Blake	ADULT	Web Design Guide	Technology	1	100

Completed queries

Overdue Loans

Show loans returned after the due date or still overdue. Include member, book and days late.

Criterion: FineDue: [DaysLate]*[DailyFine]

Member Loans

Join Member, Book and Loan and show all borrowing details sorted by LastName.

Calculation: No additional calculation

Completed form

Member Information Form with Loan subform and clear borrowing headings.

Completed report

Overdue Report grouped by MemberType and showing count and total FineDue.

Final section check

- Every question in the practice brief has a visible completed result.
- The output uses the same project facts as the other sections.
- The work has been tested or checked rather than assumed correct.
- This SPARK example is watermarked and is for study only. Build your own teacher-assigned SBA.

SPARK REFERENCE - DO NOT SUBMIT

SPARK REFERENCE - DO NOT SUBMIT

SPARK REFERENCE - DO NOT SUBMIT

SPARK REFERENCE - DO NOT SUBMIT